

HARSH SHARMA

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SUMMARY

Backend & Infrastructure Engineer with **3 years** of experience building scalable systems, APIs, and developer tools across **AI, EDA, Robotics, and Web3**. Deep expertise in **Python, Go, TypeScript**, and distributed backend architectures with hands-on experience in **Docker, Kubernetes, cloud infrastructure, and agentic AI integration**. Proven record of **end-to-end technical ownership**, cross-functional leadership, and founding-team collaboration at multiple startups. Passionate about building **simulation environments and infrastructure for AI agent training**.

EXPERIENCE

Aurora System (csfirst) — Backend & Automation Engineer, Independent Contributor

Hybrid (Bangalore) | Nov 2024 – Present | Direct report to Founder; sole India-based engineer (team across USA & China)

- **Agentic AI Extraction Pipeline:** Built a multi-stage agentic pipeline combining web-scraping, LLM-driven analysis, and vision models to autonomously extract pin maps, power rails, and metadata from 1000s of semiconductor PDFs (Qualcomm project). Orchestrated agent workflows with retry logic, validation, and human-in-the-loop fallback.
- **Parser Performance Optimization:** Re-architected core AuroraDB parser with hash-map indexing and lazy evaluation, achieving **98% latency reduction** (2-hour jobs completed in under 1 minute). Profiled and eliminated critical bottlenecks in the translation pipeline.
- **EDA Format Translation Engine:** Designed and built robust translators converting AuroraDB into industry-standard formats (Ansys AEDB, Cadence SPD, AAF, IPC-2581), enabling seamless interoperability across PowerSI, Sigrit, and Ansys EDT. Architected shared translation framework serving 5+ output formats from a single source.
- **AuroraDB & EnhancedAuroraDB:** Designed the structured database format for EDA workflows with full CRUD APIs. Extended into EnhancedAuroraDB supporting customer-specific re-routing and design modifications. Built GenAI-powered synthetic test case generator for large-scale regression validation.

Orangewood Labs — Team Lead & System Software Engineer (R&D), later Technical Advisor

Remote | Sep 2024 – May 2025 | Led engineering efforts directly with co-founders; represented company in customer & partner meetings

- **Technical Leadership:** Sole System Software Engineer and Team Lead for robot control platform. Elevated to **Technical Advisor**, attending customer demos and partnership meetings with leading robotics companies to close deals and share technology.
- **Agentic Automation & Control:** Developed AI-driven control algorithms (moveCircle, moveProcess, moveTrajectory) with autonomous error recovery and adaptive path planning. Implemented point cloud to robotic pose mapping with sensor-to-robot frame transformations for motion planning.
- **Performance Optimization:** Optimized Python and C++ control modules, cutting robotic arm runtime latency by **30%**. Built automation ecosystem with custom debug tooling for Robot Operating System stability.

Cadence Design Systems — Software Development Engineer (R&D)

On-site (Noida) | Aug 2023 – Aug 2024

- Built **Remote Tempus Session (RTS)** feature enabling decentralized, shared workspaces for large-loaded EDA designs, boosting team productivity and remote debugging. Integrated CI/CD automation with IBM Cloud for compilation and test orchestration.
- Engineered memory-swap optimization for design reloads. Made tool compatible with containerized Apple (NY) Docker systems using dual PID/UUID file management. Delivered customer-requested enhancements in PBA and GBA (Static Timing Analysis).

Ansys Software Pvt. Ltd. — R&D Intern

Hybrid (Noida) | Jan 2023 – Jul 2023

- Replaced legacy Terra parser with Verific Parser across **20,000+ lines**, reducing build time by **10%**. Built debugging tooling with GDB and productivity scripts for faster regression testing and analysis.

GoGlocal — Team Lead, Full-Stack Development

Mumbai | Apr 2022 – Jul 2022 | Led development team; worked directly with founding members

- Led development of global e-commerce platform (React, Django, REST, AWS EC2/RDS) with multi-country marketplace integration (Amazon, eBay), international payments, and B2B expansion features.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, C/C++, Bash, Java, CUDA, Kotlin | **Backend:** FastAPI, Django, gRPC, REST, Node.js

Infrastructure: Docker, Kubernetes, AWS (EC2/S3/RDS), Azure, Jenkins, CI/CD | **Databases:** PostgreSQL, MongoDB, Redis, MySQL, Snowflake

AI/ML & Agentic: LangChain, PyTorch, TensorFlow, OpenAI APIs, RAG, LLM Fine-tuning, Prompt Engineering, Agentic Pipelines, GenAI Integration

Tools & Protocols: Playwright, Git, Linux, gRPC, WebSocket, ROS, GDB, JIRA, Vim | **Formats:** AEDB, SPD, AAF, IPC-2581

EDUCATION

Netaji Subhas University of Technology (NSUT) — B.Tech, Electrical Engineering | Delhi, India | GPA: 8.4/10 | May 2023

Relevant: Neural Networks & Fuzzy Logic, Computer Networks, Data Structures & Algorithms, Big Data Science & Tools, Data Warehouse Management

LEADERSHIP & RECOGNITION

- **Founding-team collaborator** at 3 startups (GoGlocal, Orangewood Labs, Aurora System) -- direct access to co-founders, strategic input on product direction
- **IEEE Outstanding Volunteer Award** (Delhi Section, 2022) -- organized hackathons, workshops, mentored juniors at NSUT IEEE Chapter
- **Languages:** English (Professional), Hindi (Native), German (Intermediate), Chinese (Basic)